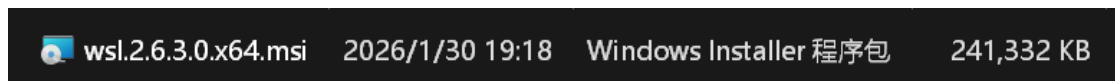


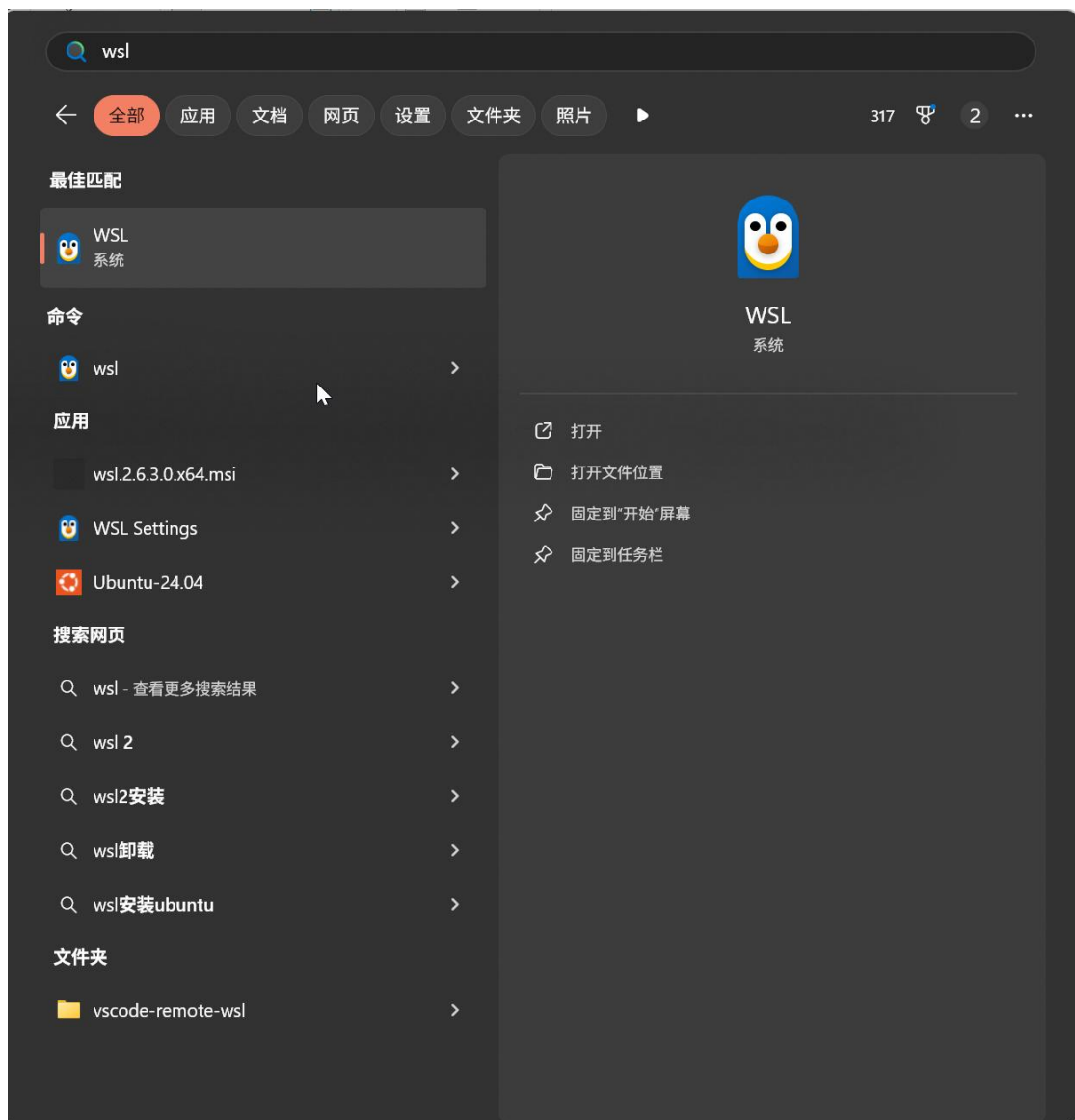
## 一、安装 wsl

双击 wsl.2.6.3.0.x64, 如图文件



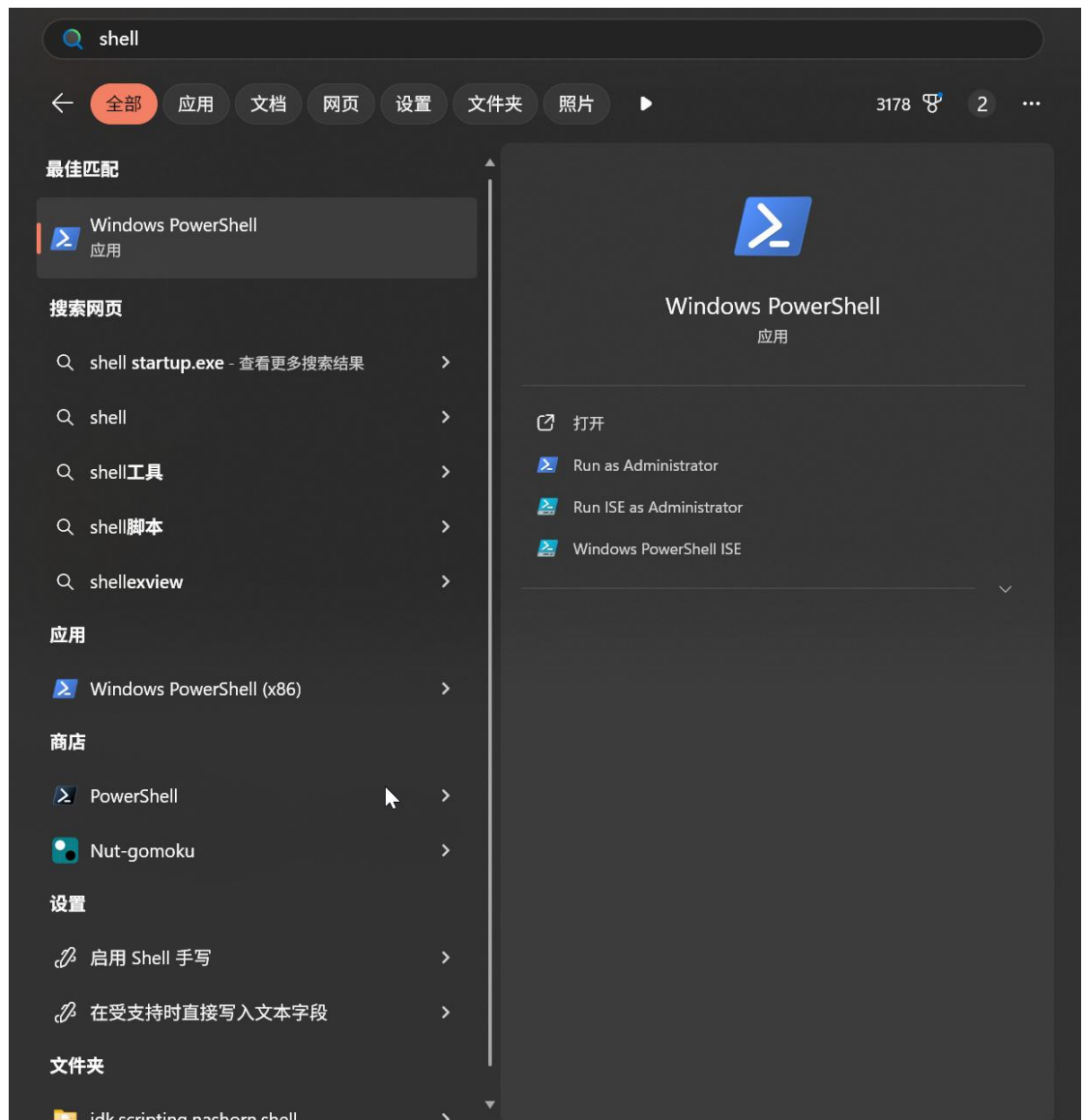
正常来说双击后就会自动完成安装 wsl

如果想检查一下是否安装成功, 按住 <win> (空格左边的第二个键一般就是 win 键) 再按 q 打开搜索界面, 输入 wsl, 出现下图, 表示安装成功了

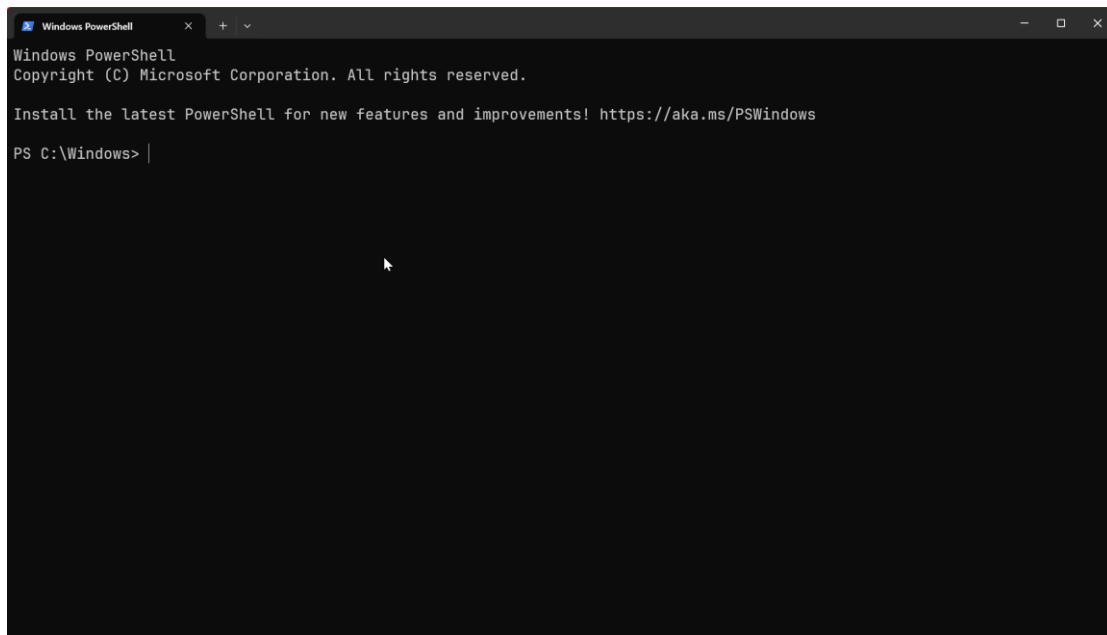


## 二、安装 Ubuntu

1、进入 powershell(可以直接搜索 shell 然后进入)



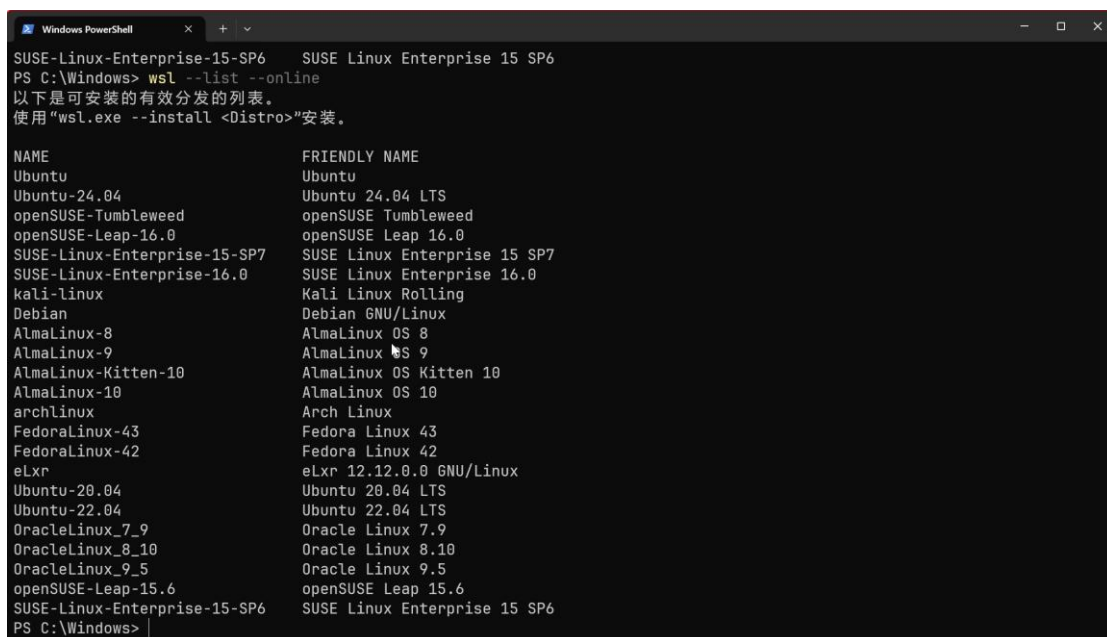
进入 powershell 之后应该就是如下图



## 2、查看并安装发行版

使用命令 `wsl --list --online` 可以查看有哪些发行版，如果不想看可以跳过这一步

如下图



使用命令 `wsl.exe --install Ubuntu-24.04` 进行安装

使用命令 `wsl -l -v` 可以查看当前安装好的版本状态

```
PS C:\Windows> wsl -l -v
  NAME                STATE      VERSION
* Ubuntu-24.04        Stopped    2
```

使用命令 `wsl` 可以直接进入 Ubuntu

```
PS C:\Windows> wsl
robocup@laptop-lr:/mnt/c/Windows$ |
```

以下操作都将在 Ubuntu 中进行

先使用 `cd` 返回到根目录

```
robocup@laptop-lr:~$ |
```

蓝色波浪号代表根目录

### 三、安装监视器

使用命令

`wget`

<https://github.com/magmaOffenburg/RoboViz/releases/download/2.0.1/RoboViz.tar.gz>

[Viz.tar.gz](https://github.com/magmaOffenburg/RoboViz/releases/download/2.0.1/RoboViz.tar.gz)(网址和 `wget` 都要在同一行)

```
robocup@laptop-lr:~$ wget https://github.com/magmaOffenburg/RoboViz/releases/download/2.0.1/RoboViz.tar.gz
```

[illegible]

然后会得到一个 bin 文件如图

```
robocup@laptop-lr:~$ ls
RoboViz.tar.gz
robocup@laptop-lr:~$ tar -zxvf RoboViz.tar.gz
bin/
bin/CHANGELOG.md
bin/roboviz.bat
bin/roboviz.sh
bin/LICENSE.md
bin/config.txt
bin/RoboViz.jar
bin/NOTICE.md
robocup@laptop-lr:~$ ls
RoboViz.tar.gz  bin
```

先使用命令 `sudo apt update` 更新软件包列表(会要求你输入密码)

然后使用命令 `sudo apt install openjdk-17-jdk -y` 安装 java

可以使用 `java -version` 查看是否安装好了

```
robocup@laptop-lr:~$ java --version
openjdk 17.0.17 2025-10-21
OpenJDK Runtime Environment (build 17.0.17+10-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 17.0.17+10-Ubuntu-124.04, mixed mode, sharing)
```

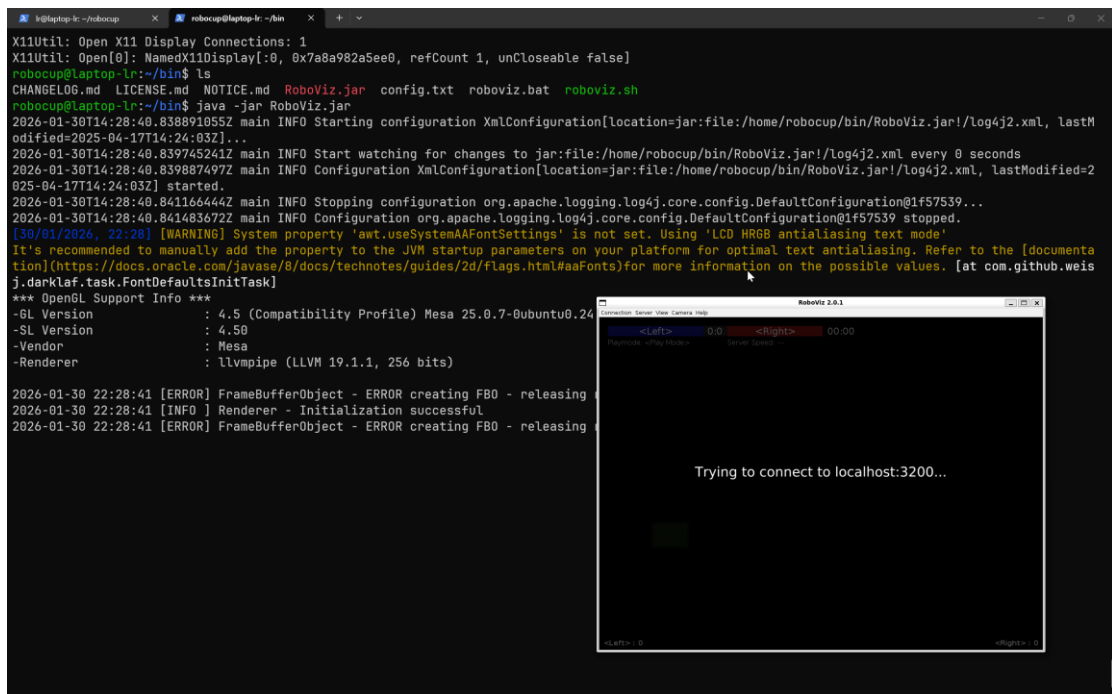
然后在 bin 目录下就可以启动 RoboViz 了 (`cd ~/bin` 可以切换到 bin 目录)

`Java -jar RoboViz.jar`

或

`./roboviz.sh`

都可以启动



The screenshot shows a terminal window with the following output:

```
X11Util: Open X11 Display Connections: 1
X11Util: Open[0]: NamedX11Display[:0, 0x7a8a982a5ee0, refCount 1, unCloseable false]
robocup@laptop-lr:~/bin$ ls
CHANGELOG.md LICENSE.md NOTICE.md RoboViz.jar config.txt roboviz.bat roboviz.sh
robocup@laptop-lr:~/bin$ java -jar RoboViz.jar
2026-01-30T14:28:40.838891055Z main INFO Starting configuration XmlConfiguration[location=jar:file:/home/robocup/bin/RoboViz.jar!/log4j2.xml, lastModified=2025-04-17T14:24:03Z]...
2026-01-30T14:28:40.839745241Z main INFO Start watching for changes to jar:file:/home/robocup/bin/RoboViz.jar!/log4j2.xml every 0 seconds
2026-01-30T14:28:40.839887497Z main INFO Configuration XmlConfiguration[location=jar:file:/home/robocup/bin/RoboViz.jar!/log4j2.xml, lastModified=2025-04-17T14:24:03Z] started.
2026-01-30T14:28:40.841166444Z main INFO Stopping configuration org.apache.logging.log4j.core.config.DefaultConfiguration@1f57539...
2026-01-30T14:28:40.841483672Z main INFO Configuration org.apache.logging.log4j.core.config.DefaultConfiguration@1f57539 stopped.
[30/01/2026, 22:28] [WARNING] System property 'awt.useSystemAAFontSettings' is not set. Using 'LCD HRGB antialiasing text mode'
It's recommended to manually add the property to the JVM startup parameters on your platform for optimal text antialiasing. Refer to the [documentation](https://docs.oracle.com/javase/8/docs/technotes/guides/2d/flags.html#aaFonts) for more information on the possible values. [at com.github.weisj.darklaf.task.FontDefaultsInitTask]
*** OpenGL Support Info ***
-GL Version      : 4.5 (Compatibility Profile) Mesa 25.0.7-0ubuntu0.24
-SL Version      : 4.50
-Vendor          : Mesa
-Renderer        : llvmpipe (LLVM 19.1.1, 256 bits)

2026-01-30 22:28:41 [ERROR] FrameBufferObject - ERROR creating FBO - releasing
2026-01-30 22:28:41 [INFO ] Renderer - Initialization successful
2026-01-30 22:28:41 [ERROR] FrameBufferObject - ERROR creating FBO - releasing
```

The RoboViz 2.0.3 application window is shown with the following text:

```
RoboViz 2.0.3
Connection Serial view camera help
<Left> 0:0 <Right> 00:00
Playback: stop mode> Server Speed: ---
Trying to connect to localhost:3200...
<Left>: 0 <Right>: 0
```

`<CTRL> + C` 可以退出 RoboViz

根目录下的 `RoboViz.tar.gz` 就可以删除了, 使用 `rm RoboViz.tar.gz`

#### 四、安装 simspark

接下来我们要安装 simspark 了, 建议先切换到根目录里 (`cd ~`)

使用命令 `git clone https://gitlab.com/robocup-sim/SimSpark.git`

```
robocup@laptop-lr:~$ git clone https://gitlab.com/robocup-sim/SimSpark.git
Cloning into 'SimSpark'...
remote: Enumerating objects: 40028, done.
remote: Counting objects: 100% (1758/1758), done.
remote: Compressing objects: 100% (757/757), done.
remote: Total 40028 (delta 1186), reused 1314 (delta 993), pack-reused 38270 (from 1)
Receiving objects: 100% (40028/40028), 24.78 MiB | 6.89 MiB/s, done.
Resolving deltas: 100% (30384/30384), done.
robocup@laptop-lr:~$ ls
RoboViz.tar.gz  SimSpark  bin
robocup@laptop-lr:~$ |
```

通过 `cd SimSpark` 进入 Simspark

```
robocup@laptop-lr:~/SimSpark$ |
```

然后我们需要先安装 `cmake` 工具

使用命令 `sudo apt install cmake` 安装 `cmake`（安装过程中会问你是否继续，按 `Y` 就可以了）

然后我们需要安装编译 `SimSpark` 所需要的依赖

使用以下命令：

```
sudo apt update && sudo apt install -y cmake build-essential git libboost-all-dev
ruby-dev libode-dev libsdl1.2-dev libdevil-dev libfreetype6-dev libxml2-dev libtbb-
dev libqt5opengl5-dev qtbase5-dev
```

```
robocup@laptop-lr:~$ sudo apt update && sudo apt install -y cmake build-essential git libboost-all-dev ruby-dev libode-dev libsdl1.2-dev libdevil-d
ev libfreetype6-dev libxml2-dev libtbb-dev libqt5opengl5-dev qtbase5-dev
```

然后在 `simspark` 目录下启动编译脚本

`./build.sh`（也会让你输一次密码）

可以检查下，如果 `rcssserver3d` 文件夹和 `spark` 文件夹下都有 `build` 文件就说明是编译成功了

```

robocup@laptop-lr:~/SimSpark$ ls
build.sh clean.sh rcssserver3d rsedit simspark-utilities spark
robocup@laptop-lr:~/SimSpark$ cd rcssserver3d/
robocup@laptop-lr:~/SimSpark/rcssserver3d$ ls
AUTHORS      ChangeLog  README      RELEASE     build      data      linux      rcssagent3d  rcssserver3d_config.h.cmake
CMakeLists.txt  INSTALL  README-soccer  THANKS      cmake      doc      macosX      rcssmonitor3d  sparkgui
COPYING        NEWS      README.MacOSX  TODO        cmake_uninstall.cmake.in  guiplugin  plugin      rcssserver3d  windows
robocup@laptop-lr:~/SimSpark/rcssserver3d$ cd ..
robocup@laptop-lr:~/SimSpark$ cd spark/
robocup@laptop-lr:~/SimSpark/spark$ ls
AUTHORS      ChangeLog  README  TODO  cmake      doc      lib      spark      utility
CMakeLists.txt  INSTALL  RELEASE  build  cmake_uninstall.cmake.in  guilayout  macosX  sparkconfig.h.cmake  windows
COPYING        NEWS      THANKS  carbon  data      guiplugin  plugin  test
robocup@laptop-lr:~/SimSpark/spark$

```

接下来就可以看到球场了

使用命令 `rcssserver3d` 打开服务器（在任意地方使用都可以）

然后到 bin 目录里启动 RoboViz(方法见上面)

The screenshot shows a laptop screen with a Java application window titled "RoboViz 2.6.3". The application displays a green soccer field with a white center circle and a goal. The interface includes a "Correction Server" window with "Left" and "Right" sliders, and a "Playmode" dropdown set to "BeforeKickOff". The background shows a terminal window with Java logs and a file explorer window.

## 五、让球员上场

到根目录里

使用命令 `git clone https://github.com/m-abr/FCPCodebase.git` 下载球员代码

```
robocup@laptop-lr:~$ ls
FCPCCodebase  RoboViz.tar.gz  SimSpark  bin
robocup@laptop-lr:~$
```

FCPCodebase 就是球员代码

但是现在还不能启动球员，我们需要下载 python 和 pip 还有一些库

我们先安装 python 和 pip

使用命令 `sudo apt install python3 python3-pip`

```
robocup@laptop-lr:~$ python --version
Command 'python' not found, did you mean:
  command 'python3' from deb python3
  command 'python' from deb python-is-python3
robocup@laptop-lr:~$ pip --version
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
robocup@laptop-lr:~$ |
```

然后装 numpy 库

使用命令

`sudo apt install python3-numpy`

装系统开发库

`sudo apt install pybind11-dev`

安装 python 模块

`sudo apt install python3-pybind11`

安装 GSL

`sudo apt install libgsl-dev`

安装完以上之后就可以启动球员了

然后进入球员代码文件夹

```
robocup@laptop-lr:~$ cd FCPCCodebase/
robocup@laptop-lr:~/FCPCCodebase$ ls
LICENSE      Run_One_vs_One.py  agent      communication  logs      start.sh      start_fat_proxy_debug.sh  world
README.md    Run_Player.py      behaviors  cpp            math_ops  start_debug.sh start_penalty.sh
Run_Full_Team.py  Run_Utils.py      bundle     kill.sh        scripts   start_fat_proxy.sh  start_penalty_debug.sh
robocup@laptop-lr:~/FCPCCodebase$ |
```

使用命令 `./start.sh` 就可以在监视器上看到 11 个球员了

```
robocup@laptop-lr:~/FCPCodebase$ ls
LICENSE          Run_One_vs_One.py  agent      communication  kill.sh      scripts      start_fat_proxy.sh  start_penalty_debug.sh
README.md        Run_Player.py      behaviors  config.json    logs         start.sh     start_fat_proxy_debug.sh  world
Run_Full_Team.py Run_Utils.py       bundle     cpp            math_ops     start_debug.sh  start_penalty.sh

robocup@laptop-lr:~/FCPCodebase$ ./start.sh
robocup@laptop-lr:~/FCPCodebase$ Connected agent 6 ('127.0.0.1', 45688)
Connected agent 4 ('127.0.0.1', 45698)
Connected agent 7 ('127.0.0.1', 45704)
Connected agent 3 ('127.0.0.1', 45708)
Connected agent 2 ('127.0.0.1', 45720)
Connected agent 1 ('127.0.0.1', 45732)
Connected agent 9 ('127.0.0.1', 45734)
Connected agent 5 ('127.0.0.1', 45736)
Connected agent 11 ('127.0.0.1', 45748)
Connected agent 8 ('127.0.0.1', 45764)
Connected agent 10 ('127.0.0.1', 45776)
```



## 六、让他们踢起来

我们先回到根目录

```
robocup@laptop-lr:~$ ls
FCPCodebase  RoboViz.tar.gz  SimSpark  bin
robocup@laptop-lr:~$ |
```

再复制一份球员代码

使用命令 `cp -r FCPCodebase FCPCodebase_backup`

```
robocup@laptop-lr:~$ cp -r FCPCodebase FCPCodebase_backup
robocup@laptop-lr:~$ ls
FCPCodebase  FCPCodebase_backup  RoboViz.tar.gz  SimSpark  bin
robocup@laptop-lr:~$ |
```

再进入到 `FCPCodebase_backup` 目录下，使用 `nano start.sh` 打开 `start.sh`

```
robocup@laptop-lr:~$ ls
FCPCodebase  FCPCodebase_backup  RoboViz.tar.gz  SimSpark  bin
robocup@laptop-lr:~$ cd FCPCodebase_backup/
robocup@laptop-lr:~/FCPCodebase_backup$ ls
LICENSE          Run_One_vs_One.py  agent      communication  kill.sh      scripts      start_fat_proxy.sh  start_penalty_debug.sh
README.md        Run_Player.py      behaviors  config.json    logs         start.sh     start_fat_proxy_debug.sh  world
Run_Full_Team.py Run_Utils.py       bundle     cpp            math_ops     start_debug.sh  start_penalty.sh

robocup@laptop-lr:~/FCPCodebase_backup$ nano start.sh
```

下图白色选中的是队伍名，我们需要更改队伍名，可以随便改，只有两个队伍名不一样就可以了

```
GNU nano 7.2
#!/bin/bash
export OMP_NUM_THREADS=1

host=${1:-localhost}
port=${2:-3100}

for i in {1..11}; do
    python3 ./Run_Player.py -i $host -p $port -u $i -t FCPortugal -P 0 -D 0 &
done
```

```
GNU nano 7.2 start.sh
#!/bin/bash
export OMP_NUM_THREADS=1

host=${1:-localhost}
port=${2:-3100}

for i in {1..11}; do
    python3 ./Run_Player.py -i $host -p $port -u $i -t FCPortugal_buckup -P 0 -D 0 &
done
```

然后使用 <CTRL> + X 退出 nano

它会问你是否保存 按 y 然后按 回车就可以了

然后使用 ./start.sh 如下图启动球员

```
robocup@laptop-lr:~/FCPCodebase_backup$ ./start.sh
robocup@laptop-lr:~/FCPCodebase_backup$ Connected agent 2 ('127.0.0.1', 45324)
Connected agent 5 ('127.0.0.1', 45334)
Connected agent 6 ('127.0.0.1', 45340)
Connected agent 1 ('127.0.0.1', 45350)
Connected agent 7 ('127.0.0.1', 45366)
Connected agent 10 ('127.0.0.1', 45370)
Connected agent 9 ('127.0.0.1', 45368)
Connected agent 4 ('127.0.0.1', 45394)
Connected agent 11 ('127.0.0.1', 45404)
Connected agent 3 ('127.0.0.1', 45406)
Connected agent 8 ('127.0.0.1', 45418)
```

正常情况下监视器如图



按 k 就可以开球了，球员就会开始踢球

如果出现不正常情况可以试着重启服务器（`pkill -9 rcssserver3d` 可以强制关闭服务器）或者重启 wsl（在 powershell 里 `wsl --shutdown` 可以关机 wsl）



祝各位好运，有问题随时欢迎来问